

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A) | Date: 25-03-26
Time: 3 Hrs | Max. Marks: 300

25-03-26_SR.IIT_STAR CO-SC(MODEL-A)_JEE MAIN_SPCL GTM-13_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

Rankers Academy JEE
THE NARAYANA GROUP

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PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. In the expression $\alpha = \frac{Fv^2}{\beta^2}\log_e\left(\frac{2\pi\beta}{v^2}\right)$, find the dimensions of α and β. Here F is the force and v is the velocity.

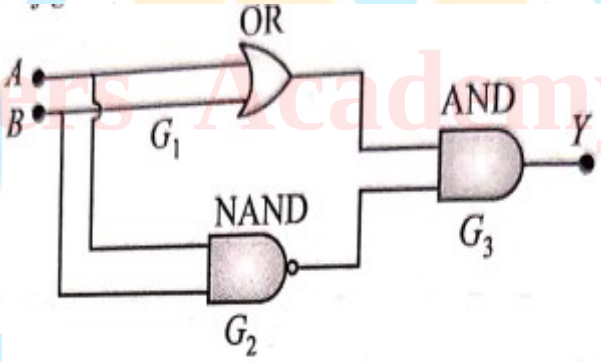
A) $\left[M^1L^{-1}T^0\right],\left[L^2T^{-2}\right]$ B) $\left[L^2T^{-2}\right],\left[ML^{-1}T^0\right],$

C) $\left[M^1L^{-2}T^1\right],\left[M^1L^1T^{-1}\right]$ D) $\left[M^{-1}L^1T^0\right],\left[L^{-2}T^2\right]$

2. The circular head of a screw gauge is divided into 200 divisions and move 1 mm ahead in one revolution. If the same instrument has a zero error of -0.05 mm and the reading on the main scale in measuring diameter of a wire is 6 mm and that on circular scale is 45, find the diameter of the wire.

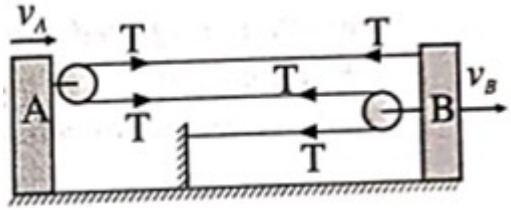
A) 6.725mm B) 6.275mm C) 6.752mm D) 6.572mm

3. The following configuration of gate is equivalent to



A) NAND B) XOR C) OR D) AND

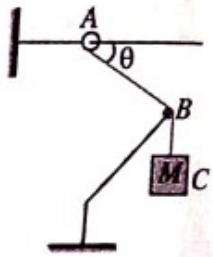
4. In the arragement shown in the figure the block B stats from rest and moves towards right with a constant acceleration. After time t the velocity of A with respect to B become v. Determine the acceleration of A.



A) $\frac{2v}{3t}$ B) $\frac{2v}{t}$ C) $\frac{3v}{t}$ D) $\frac{3v}{2t}$

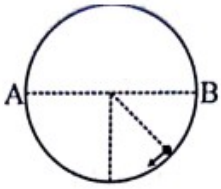
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5. A smooth ring A of mass m can slide on a fixed horizontal rod. A string tied to the ring passes over a fixed pulley B and carries a block C of mass M=2m as shown in figure. At an instant the string between the ring and the pulley makes an angle θ with the rod. Find the acceleration of the ring just after released from $\theta = 30^0$, [g=10 m/s^2]



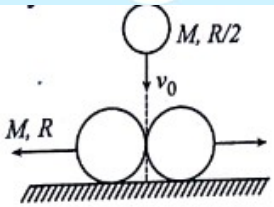
A) $5\sqrt{3}\,m/s^2$ B) $2\sqrt{3}\,m/s^2$ C) $6\,m/s^2$ D) $4\sqrt{3}\,m/s^2$

6. A particle of mass m oscillates inside a smooth spherical shell of radius R (released from point B). At any instant the kinetic energy of the particle is K. Then the force applied by the particle on the shell at this instant is:



A) $\frac{K}{R}$ B) $\frac{2K}{R}$ C) $\frac{3K}{R}$ D) $\frac{4K}{R}$

7. In the figure shown, the two identical balls each of mass M and radius R, are placed in contact with each other on the frictionless horizontal surface. The third ball of mass M and radius R/2, is coming down vertically and has a velocity $= v_0$ just before it strikes simultaneously these two balls and itself comes to rest just after collision. Then, each of the two balls will move after collision with speed:



A) $\frac{v_0}{\sqrt{3}}$ B) $\frac{v_0}{2}$ C) $\frac{v_0}{\sqrt{5}}$ D) $\frac{2v_0}{\sqrt{5}}$

8. Given a semi-circular disc of mass m and radius R. Find its MI about an axis passing through CM and parallel to its diameter.

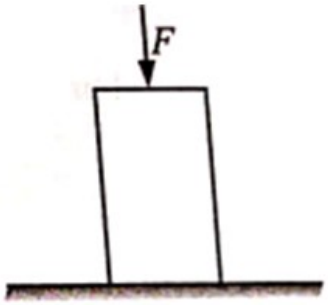
A) $mR^2\left[\frac{1}{4}-\frac{16}{9\pi^2}\right]$ B) $mR^2\left[\frac{1}{2}-\frac{16}{9\pi^2}\right]$ C) $mR^2\left[1-\frac{4}{3\pi^2}\right]$ D) $mR^2\left[2-\frac{4}{3\pi^2}\right]$

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9. If a satellite is launched from height 'h' with a speed 'u' horizontally, find the angle with the vertical at which the satellite will hit the earth`s surface, if h=R and $u=\sqrt{\frac{GM}{7R}}$.

A) 30^0 B) 90^0 C) 45^0 D) 60^0

10. A metal cylinder of length L is subjected to a uniform compressive force F as shown in the figure. The material of the cylinder has Young`s modulus Y and poisson`s ratio σ. The change in volume of the cylinder is:

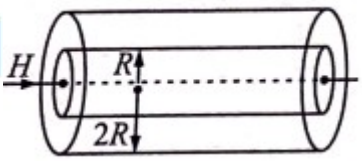


A) $\frac{\sigma FL}{Y}$ B) $\frac{(1-\sigma)FL}{Y}$ C) $\frac{(1+2\sigma)FL}{Y}$ D) $\frac{(1-2\sigma)FL}{Y}$

11. A tank filled with water (density $\rho_\omega = 1000 kg/m^2$) and oil of (density $\rho_{oil} = 900\ kg/m^3$). The height of water is 1.00 m and of the oil is 4.00 m. Find the velocity of efflux through a hole at the bottom of the tank.

A) 6.5 m/s B) 9.5 m/s C) 11.5 m/s D) 12.5 m/s

12. A cylinder of radius R made of material of thermal conductively K is surrounded by cylindrical shell of inner radius R and outer radius 2R made of a material of thermal conductively 2K. The two ends of the combined system are maintained at two different temperatures. There in no loss of heat across the cylindrical surface and system is in steady state. Find the effective thermal conductivity of the system.



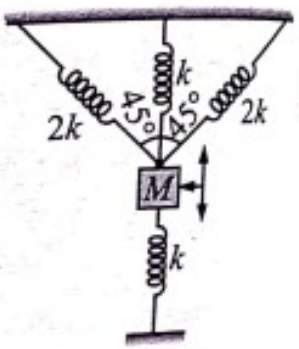
A)4K B) $\frac{4}{7}K$ C) $\frac{7}{4}K$ D) 7K

13. An ideal gas $\left(C_p/C_v=\gamma\right)$ is taken through a process in which the pressure and the volume vary as $P=aV^b$. Find the value of b for which the specific heat capacity in the process is zero.

A) $-\gamma/3$ B) $-\gamma$ C) $-\gamma/2$ D) -2γ

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14. In the device shown, find time period of small vertical oscillations of the block. All the springs are massless and block is of mass M.



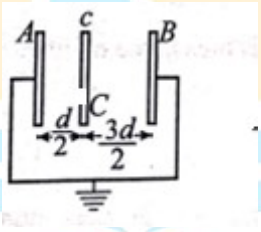
A) $3\pi\sqrt{\frac{M}{k}}$ B) $2\pi\sqrt{\frac{M}{k}}$ C) $\pi\sqrt{\frac{M}{k}}$ D) $\frac{\pi}{2}\sqrt{\frac{M}{k}}$

15. The first overtone of an open organ pipe beats with the first overtone of a close organ pipe with a beat frequency of 2.2 Hz. The fundamental frequency of the closed organ pipe is 110 Hz. Find the lengths of the pipes. Velocity of sound in air=330 m/s

A) $L_0 = 0.99$ m, $L_C = 0.75$m B) $L_0 = 1.01$ m, $L_C = 0.99$ m

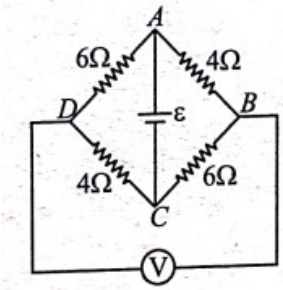
C) $L_0 = 1.00$ m, $L_C = 1.02$ m D) $L_0 = 0.99$ m, $L_C = 1.00$ m

16. A conducting plate C carrying a charge q is placed between two earthed plates as shown. Find charges on the plates A & B.



A) $\frac{q}{3}, \frac{2q}{3}$ B) $\frac{q}{2}, \frac{q}{2}$ C) $\frac{q}{4}, \frac{3q}{2}$ D) $\frac{3q}{4}, \frac{q}{4}$

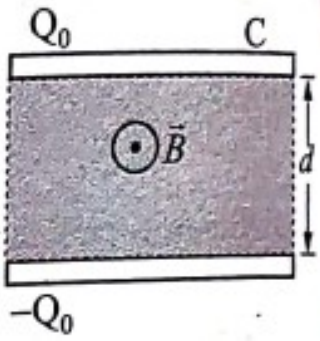
17. In the circuit shown, find reading of voltmeter of large resistance. The emf of the cell is ε.



A) $\varepsilon/5$ B) $\varepsilon/4$ C) $\varepsilon/3$ D) $\varepsilon/2$

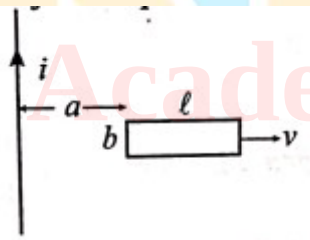
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18. A slab of resistance R is inserted between two parallel plates of a capacitor charged to Q_0. The capacitor discharged through the solid. A magnetic field is present throughout and is out of the plane of the paper. The total momentum given to the slab after complete discharge is (given that the capacitance of the capacitor is C and the plates are separated by a distance d)



A) CQ_0B B) dBQ_0 C) zero D) infinity

19. A rectangular metallic loop of length ℓ and width b is placed coplanarly with a long wire carrying a current i. The loop is moved perpendicular to the wire with a speed v in the plane containing the wire and the loop. Calculate the emf induced in the loop when the rear end of the loop is at a distance a from the wire.



A) $\frac{\mu_0 Iv}{2\pi\, ab\ell}$ B) $\frac{\mu_0 ib\ell}{2\pi av(a+\ell)}$ C) $\frac{\mu_0 i(b\ell v)}{2\pi a(a+\ell)}$ D) $\frac{\mu_0 iv}{a(a+\ell)}$

20. A point charge is moving along a straight line with a constant velocity v. Consider a small area A perpendicular to the direction of motion of the charge. Find displacement current through the area when it is at a distance x from the charge. (The value of x is not very large)

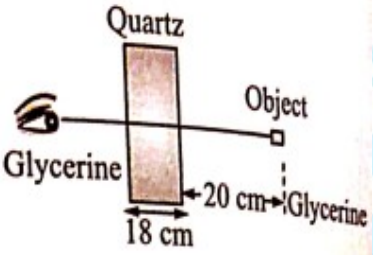
A) $\frac{2qAv}{\pi x^3}$ B) $\frac{qAv}{2\pi x^3}$ C) $\frac{qAv}{2\pi \in_0 x^3}$ D) $\frac{qAv}{4\pi x^3}$

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

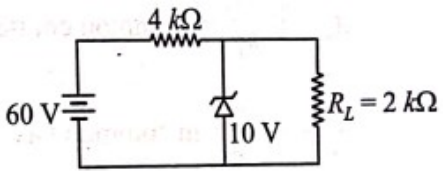
21. Given that, velocity of light in quartz $=1.5\times10^{8}$ m/s and velocity of light in glycerin $=\frac{9}{4}\times10^{8}$ m/s. Now a slab made of quartz, is placed in glycerin as shown. The shift produced by the slab is ____x____ cm. Find x.



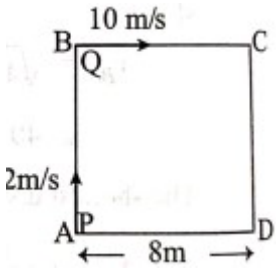
22. In YDSE set-up d=1 mm, $\lambda = 600$ nm and D=1 m. The slits produce same intensity on the screen. The minimum distance between two points on the screen having 75% of the maximum intensity is $\alpha\times10^{-4}$ m, Find α _____

23. A hydrogen atom at rest emits a photon of wavelength 122 nm. The recoil speed of the hydrogen atom is $x\times10^{-2}$ m/s, [$h = 6.6\times10^{-34}$ Js mass of proton $=1.67\times10^{-27}$ kg],
Find x_____

24. A Zener diode is connected to a battery and a load as shown below. The current in diode is $\alpha\times10^{-4}$ A, Then α is_____



25. Two men P and Q are standing at corners A and B of square ABCD of side 8m. They start moving along the track with constant speed 2 m/s and 10 m/s respectively. The time when they will meet for the first time is_____ s.



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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. The latent heat of vaporization of substance is 540 cal/g and if the Boiling point is 100^0C, then its ebullioscopic constant is

A) $0.513\,Kkg^{-1}mol^{-1}$ B) $1.026\,Kkg^{-1}mol^{-1}$ C) $1.832\,Kkg^{-1}mol^{-1}$ D) $1.86\,Kkg^{-1}mol^{-1}$

27. Which of the following statements are correct?

1) Units of second order reaction is $mol^{-1}lits^{-1}$

2) Decomposition of HI on gold surface is zero order reaction

3) for first order $\ell n[R]$ vs t Graph gives a straight line with slope of -k

([R]= conc. of Reactant R after time t)

4) Hydrogenation of ethene is an example of second order Reaction

Choose the correct answer from the options given below

A) 1,2 Only B) 1,2,3 & 4 Only C) 2,3 & 4 Only D) 1,2,3 Only

28. **Statement I:** Rhamnose is a 6-deoxy hexose sugar

Statement II: Rhamnose does not fit into general formula of sugars $C_n(H_2O)_n$

A) Statement-I & II both are true

B) Statement-I & II both are false

C) Statement-I is true but Statement-II is false

D) Statement-I is false but Statement-II is true

29. Which of the following statements are correct?

1) Actinides shows variable oxidations states

2) In Actinides Np and Pu elements shows maximum oxidation state

3) Trivalent Gadolinium ion possess stable Half-filled electronic configuration

4) Tetravalent Cerium ion act as strong oxidant and it can oxidize water

Choose the correct answer from the options given below

A) 2, 3 & 4 Only B) 1,2 & 3 Only C) 1,2 Only D) All are true

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30. Match the following

	List-I		List-II
1)	Photography	A)	Rhodium Complex
2)	Hardness of water	B)	$\left[Ag(S_2O_3)_2\right]^{-3}$
3)	Cis-platin	C)	Na_2EDTA
4)	Wilkinson`s catalyst	D)	Growth of tumours

A) 1 – B 2 – C 3 – A 4 – D B) 1 – B 2 – A 3 – D 4 – C

C) 1 – B 2 – C 3 – D 4 – A D) 1 – C 2 – A 3 – B 4 – D

31. At 320 K, a gas A_2 is 20 % dissociated to $A(g)$. The standard free energy change at 320 K and 1 atm in J mol^{-1} is approximately (R=8.314 J K^{-1} mol^{-1} $\log 6 = 0.778$)

A) 1844 B) 2068 C) 4281 D) 4767

32. What is the solubility product of a saturated solution of Ag_2CrO_4 in water at 298 K if the EMF of the cell Ag/Ag^+ (Saturated Ag_2CrO_4)// $Ag^+(0.1M)/Ag$ is 0.162 at 298 K

(Given $2.303\frac{RT}{F} = 0.06$, $\log 2 = 0.3$)

A) 2×10^{-4} B) 3.2×10^{-11} C) 8×10^{-12} D) 4×10^{-12}

33. If λ be the Debroglie wavelength of a thermal neutron at 27^0C, then the wavelength of the same neutron at 927^0C is

A) λ B)0.5 λ C)2 λ D)0.25 λ

34. Enthalpy of neutralization of oxalic acid is -25.4 K.Cal/mol Using strong Base NaOH. Enthalpy Change for the process $H_2C_2O_4(aq) \rightarrow 2H^+(aq) + C_2O_4^{-2}(aq)$ is

(in K.Cal)

A) 2 B)11.7 C)-11.7 D)1

35. **Statement-I:** The number of paramagnetic species among $O_2, N_2, CN^-, B_2, F_2, O_2^+$ are three.

Statement-II: Among the species $O_2, N_2, CN^-, B_2, F_2, O_2^+, N_2$ & CN^- have highest bond order.

A) Statement-I & II both are true

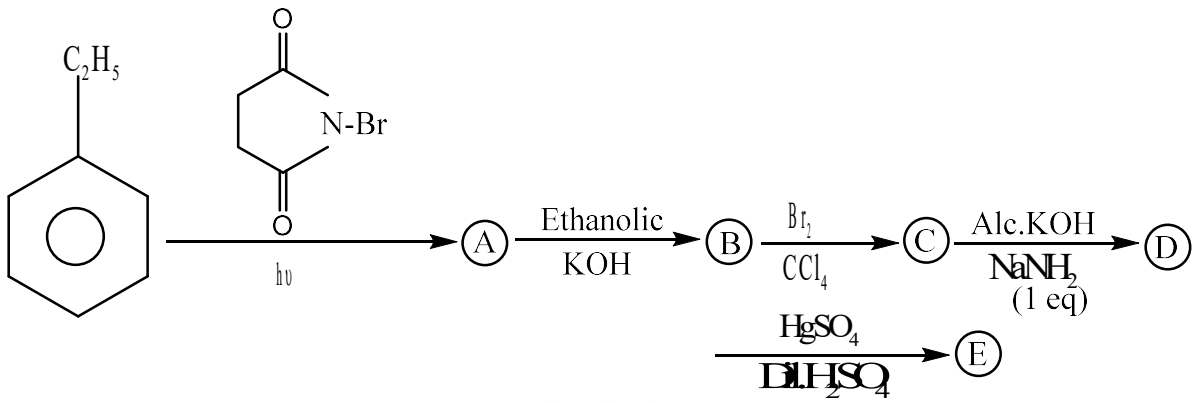
B) Statement-I & II both are false

C) Statement-I is true but Statement-II is false

D) Statement-I is false but Statement-II is true

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36. Mr. $(AN)_2$ D.Barre worked out following reaction sequence



Consider the above reaction:

(a) Number of sp^2 C-atoms in 'B' is '8'

(b) When 'D' is treated with Na/liq.ND_3 it gives trans alkene

(c) When 'E' is treated with I_2 / OH^- to give yellow precipitate

(d) Double bond equivalent (D.B.E) of 'D' is 5

(e) When D is treated with $D_2 / Pd - BaSO_4$ it gives alkene

Identify the correct statement(s), choose the correct answer from the options given below

A) a, b and e only

B) a, c and e only

C) a, b, c and e only

D) a, c, d and e only

37. Match the List-I with List-II

List-I	**List-II**
(P) $Ag(NH_3)_2OH$	(1) Birch reduction
(Q) $Na / NH_3(\ell)$	(2) Gattermann's Koch reaction
(R) $CO + HCl + AlCl_3$ anhydrous	(3) Wolf-Kishner reduction
(S) N_2H_4 / OH^-	(4) Tollen's reagent

A) P-4, Q-1, R-3, S-2

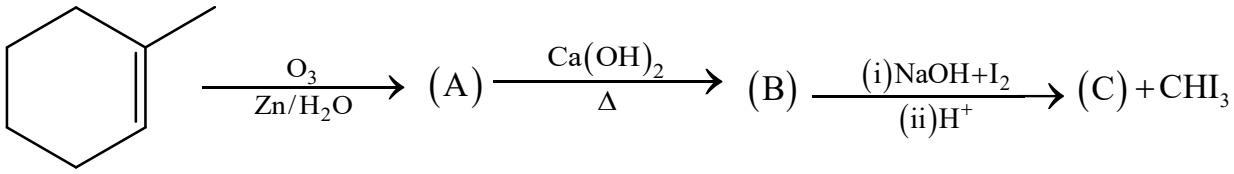
B) P-4, Q-2, R-3, S-1

C) P-4, Q-1, R-2, S-3

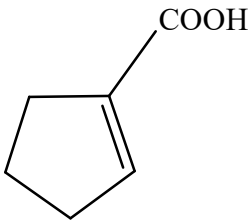
D) P-4, Q-3, R-2, S-1

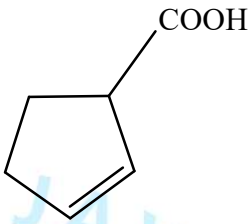
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38.

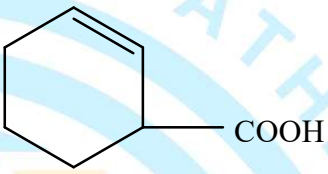


Identify product (C).

A) 

B) 

C) 

D) 

39. Statement-I: Cl_2 reacts with isobutane in $h\upsilon$ to produce 3^0 butyl chloride as major product.

Statement-II: 3^0 free radical is more stable than 1^0 free radical

A) Statement-I & II both are true

B) Statement-I & II both are false

C) Statement-I is true but Statement-II is false

D) Statement-I is false but Statement-II is true

40. The second period element which forms most acidic oxide is "A" the element "A"

i) exist as diatomic molecule

ii) has seven valence electrons

iii) has five valence electrons

iv) react with Li to form water soluble compound

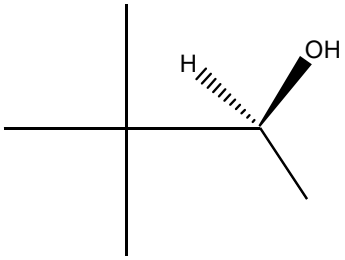
v) forms binary hydride

Correct statement(s) is / are

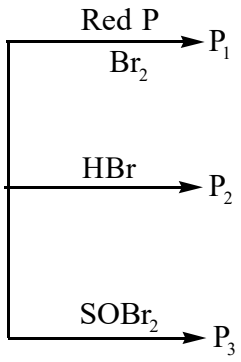
A) i, ii, iv, v only B) i, iii, iv, v only C) i, ii, iv, v only D) i, ii, v only

For More Join: @IITJEE_Advanced

41. Select incorrect statement about the products (major).



S-3,3-dimethylbutan-2-ol



A) P_1 and P_3 are identical

B) P_1 and P_2 are positional isomers

C) P_1 and P_3 are enantiomers

D) P_2 and P_3 are positional isomers

42. 0.002 grams of an organic compound has treated according to Kjeldhal`s method 0.2×10^{-4} mol of H_2SO_4 was required to neutralized ammonia the percentage of nitrogen is $7\times x\%$. What is the "x"?

A) 28 B) 7 C) 2 D) 4

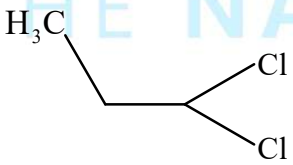
43. A compound with molecular formula $C_6H_{14}O_4$ does not give litmus test and does not give colour with 2,4 – DNP. It reacts with excess MeCOCl to give a compound whose vapour density is 117. Compound A contains how many hydroxyl groups?

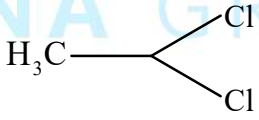
A) 1 B) 3 C) 2 D) 4

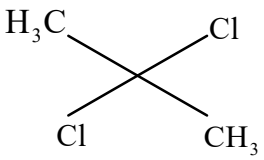
44. A light green colored salt soluble in water gives black precipitate on passing H_2S which dissolves readily in HCl. The metal ion present is:

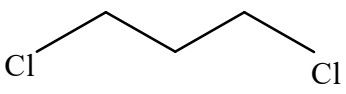
A) Co^{+2} B) Fe^{+2} C) Ni^{+2} D) Mn^{+2}

45. Which of the following compounds on hydrolysis produces a compound which can react with semi-carbazide, can reduce Tollen`s reagent and can also give positive iodoform test?

A) 

B) 

C) 

D) 

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. Consider the following equilibrium in a closed vessel $PCl_5 \rightarrow PCl_3 + Cl_2$

PCl_5 gas at a certain pressure is introduced in the container at 27^0C, however the total pressure at equilibrium at 207^0C was found to be double the initial value. Find percentage degree of dissociation at 207^0C.

47. 12 ml of a gaseous mixture of Hydrocarbons C_nH_{2n} and C_nH_{2n+2} (n is same for both) required exactly 57 ml of O_2 for combustion. If the volume of CO_2 obtained under similar conditions is 36ml, the volume ratio of hydrocarbons in the given mixture is x:y. Then x + y is_____

48. Three Complexes are given below

A) $[NiCl_4]^{-2}$ B) $[Ni(CN)_4]^{-2}$ C) $[CoF_6]^{-3}$

The magnetic moment of complexes A, B, C are x, y and z respectively. Then $z-(x+y)$ is_____ (nearest Integer)

49. In a molecule (AB) dipole moment is 3.3D then the bond length of A & B is $1.2\,\overset{0}{A}$. What is the percentage of Ionic character? (Nearest integer)

50. How many monobromo compounds will give n-butene on treatment with 'Mg' dry ether followed by H_2O? (Including stereo isomers)

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. The function $f:R\to R$ is defined by $f(x)=\frac{x}{1+|x|}$, then f is:

A) one-one and onto B) onto but not one-one

C) one-one but not onto D) neither one-one nor onto

52. For all values of $\theta\in\left(0,\frac{\pi}{2}\right)$, the determinant of the matrix $\begin{bmatrix}-2 & \tan\theta+\sec^2\theta & 3\\ -\sin\theta & \cos\theta & \sin\theta\\ -3 & -4 & 3\end{bmatrix}$ always lies in the interval:

A) $[3,5]$ B) $(4,6)$ C) $\left(\frac{5}{2},\frac{19}{4}\right)$ D) $\left[\frac{7}{2},\frac{21}{4}\right]$

53. Let A be the set of all 3-digit natural numbers and $B=\{x\in A:\text{H.C.F.}(x,15)=1\}$. Then the number of elements in B is:

A) 420 B) 360 C) 480 D) 240

54. Let $A(3,4),B(5,-2)$ and $P(\alpha,\beta),\alpha\beta\neq 0$, be three points such that PA = PB and the area of ΔPAB is 10. Then the distance of the point $Q(2\alpha-5\beta,\alpha-\beta^2)$ from the line having intercepts 3 and 1 on the x-axis and y-axis respectively, is:

A) 10 B) $\sqrt{10}$ C) 15 D) 2

55. The values of x, for which the vectors $\vec{a}=(x^2-16)\hat{i}+4\hat{j}-(x^2-25)\hat{k}$ make acute angles with the (positive directions of) coordinate axes lie in:

A) $[-5,-4]\cup[4,5]$ B) $[-5,4)\cup(-4,5]$

C) $(-5,-4)\cup(4,5)$ D) $(-5,4)\cup(-4,5)$

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56. Let $b_1 = 3, b_2, b_3, \ldots\ldots$ be a geometric progression of increasing positive numbers.

Let $\sum_{n=1}^{20} b_{3n} = 4\sum_{n=1}^{20} b_{3n-2}$. Then, the sum of the first ten terms of the G.P. is:

A) 1023 B) 2046 C) 3069 D) 3149

57. Let $\overrightarrow{OA} = \vec{a} = \frac{1}{2}\left(\hat{i} + \hat{j} - 2\hat{k}\right)$, $\overrightarrow{OC} = \vec{b} = \hat{i} - 2\hat{j} + \hat{k}$ and $\overrightarrow{OB} = 10\vec{a} + 2\vec{b}$. Let p (in sq. units) be the area of the quadrilateral OABC and q (in sq. units) be the area of the parallelogram with $\overrightarrow{OA}$ and $\overrightarrow{OC}$ as adjacent sides, then $\frac{p}{q}$ is equal to:

A) 3 B) 4 C) 6 D) 8

58. Let y = y(x) be the solution of the differential equation $e^y dy = \left(1 + x + e^y + xe^y\right)dx$ and $y(1) = 0$. Then $y(-3)$ is equal to:

A) 0 B) $\frac{3}{2}$ C) $\log_e 2$ D) $\log_e(2) - \frac{3}{2}$

59. Two persons A and B alternately throw a pair of dice. A wins if he throws a sum of 4 before B throws a sum of 9, and B wins if he throws a sum of 9 before A throws a sum of 4. The probability that A wins if B makes the first throw is:

A) $\frac{1}{5}$ B) $\frac{2}{5}$ C) $\frac{3}{5}$ D) $\frac{4}{5}$

60. Let $f(x) = \begin{vmatrix} -\cos x & \tan x & 3\sin x \\ 1 & -3x & 2x^2 \\ x^3 & x & x^2 \end{vmatrix}$. Then the value of $\lim_{x\to 0} \frac{(1+x)f(x) - 3x\sin x}{x^3}$ is:

A) 3 B) 5 C) 7 D) 8

61. Let the ellipse $x^2 + 16y^2 = 16$ be inscribed in a rectangle whose sides are parallel to the coordinate axes, and this rectangle is inscribed in another ellipse that passes through the point (16, 0), then the equation of the outer ellipse is:

A) $x^2 + 256y^2 = 16^2$ B) $x^2 + 248y^2 = 16^2$

C) $x^2 + 240y^2 = 16^2$ D) $x^2 + 232y^2 = 16^2$

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62. Let $f(x)=\int\left(\frac{1}{\log_e x}-\frac{2}{(\log_e x)^3}\right)dx$. If $f(e)=2e$, then $f(e^2)$ is equal to:

A) $\frac{e^2}{4}$ B) $\frac{e^2}{2}$ C) $\frac{3e^2}{4}$ D) $\frac{4e^2}{3}$

63. The mean deviation about the median of the data set 4,10,6,4,13,11,19, m,8,7 where $m \geq 11$ is an integer, is 4. Then m is equal to:

A) 14 B) 15 C) 16 D) 17

64. From a point A with position vector $(p\hat{i}+p\hat{j}+p\hat{k})$, B and C are the feet of the perpendiculars drawn to the lines $\vec{r}=\hat{k}+\lambda(\hat{i}+\hat{j})(\lambda \in R)$ and $\vec{r}=-\hat{k}+\mu(\hat{i}-\hat{j})(\mu \in R)$, respectively, and $\angle BAC = 90^0$. Then the value of p is equal to:

A) -1 B) $\sqrt{2}$ C) 2 D) -2

65. If the circles $x^2+y^2-2x-8y+17=r$ and $x^2+y^2-26x-18y+234=0$ intersect at exactly one point, then the sum of all possible values of r is equal to

A) 370 b) 252 C) 324 D) 227

66. If the digits at ten`s and hundred`s places in $(11)^{2016}$ are x and y respectively, then the ordered pair (x, y) is equal to:

A) (1, 6) B) (6, 1) C) (8, 1) D) (1, 8)

67. If the line $\arg(z)=\frac{\pi}{3}$ intersects the curve $|z-2\sqrt{3}i|=2, z \in C$, at two distinct points A and B, then AB equals:

A) 1 B) 2 C) 4 D) 6

68. The equation $x^3+e^x+5=0(x \in R)$, has:

A) no solution B) exactly one solution

C) exactly two solutions D) infinitely many solutions

69. If 2α is a root of $ax^2+bx+c=0$, β is a root of $ax^2-2bx-c=0$, and the real numbers a, b, c ($a>0$), are such that $\beta<\alpha$; then a root γ of $ax^2+4bx+2c=0$ always satisfies:

A) $\alpha<\gamma<2\beta$ B) $\beta<\gamma<\alpha$ C) $\gamma<\beta<\alpha$ D) $\alpha<\gamma<\beta$

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70. Let PQ be a focal chord of the parabola $y^2 = 4x$. If the center of a circle having PQ as its diameter lies on the line $\sqrt{5}y + 4 = 0$, then the length of the chord PQ is:

A) $\frac{26}{5}$ B) $\frac{36\sqrt{5}}{5}$ C) $\frac{26\sqrt{5}}{5}$ D) $\frac{36}{5}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. Let $S = \left\{\alpha : \frac{\pi}{4} < \alpha \le \frac{7\pi}{4} \text{ and } 4^{\cos^2\alpha} + 4^{\sin^2\alpha} = 4\right\}$. Then the value of $\left(\sum_{\alpha \in S}(\cot\alpha)\right) + 999$ is equal to

72. If $S_n = \sum_{r=1}^{n} T_r = n(n+1)(n+2)(n+3)$, $K = \sum_{r=1}^{10} \frac{1}{T_r}$ then the least integer greater than or equal to $\frac{1}{K}$ is equal to

73. A code word of length 4 consists of two distinct consonants in the English alphabet followed by two digits from 1 to 9, with repetition allowed in digits. If the number of code words so formed ending with an even digit is 432k, then k is equal to:

74. If the line, $y = mx$ bisects the area of the region $\left\{(x,y) : 0 \le x \le \frac{3}{2}, 0 \le y \le 1 + 4x - x^2\right\}$, then the greatest integer less than or equal to m is equal to

75. If $f(x) = [x^2] + g(x)$, where [x] represents the greatest integer $\le x$, and

$$g(x) = \begin{cases} -2x - 3 & , \quad \frac{-3}{2} \le x < -1 \\ x & , \quad -1 \le x < 1 \\ 3 - 2x & , \quad 1 \le x \le \frac{3}{2} \end{cases}$$

. Then the greatest integer less than or equal to the value of integral $\int_{-\frac{3}{2}}^{\frac{3}{2}} f(x)\,dx$, is equal to

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